Bioinformatics algorithms

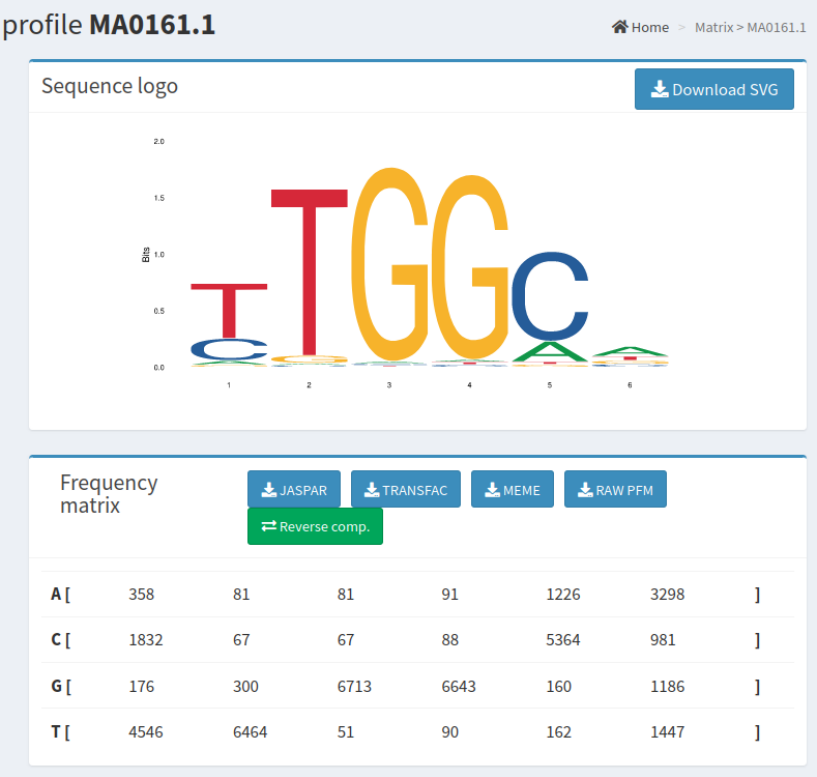
Rosario M. Piro
Politecnico di Milano

Sequence modeling and Hidden Markov Models – exercises

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1. **Alignment profile/motif**

   Consider the following motif (equivalent to an alignment profile) of the transcription factor *NFIC*:




   Note: the sequence logo and profile data are from the JASPAR database (with matrix/profile ID MA0161.1) and can be found at `https://jaspar.genereg.net/matrix/MA0161.1/`

   Answer the following questions:

   (a) This profile is given as a "Position Frequency Matrix" (PFM) which reports for every position in the alignment the number of times a specific letter was observed at that position (i.e., how many of the aligned strings contain that letter at that position). Transform the PFM into a probability matrix as we have seen it in the previous lecture:

| | Pos.1 | Pos.2 | Pos.3 | Pos.4 | Pos.5 | Pos.6 |
|---|---|---|---|---|---|---|
| A | $f_{A,1}$ | $f_{A,2}$ | $f_{A,3}$ | $f_{A,4}$ | $f_{A,5}$ | $f_{A,6}$ |
| C | $f_{C,1}$ | $f_{C,2}$ | $f_{C,3}$ | $f_{C,4}$ | $f_{C,5}$ | $f_{C,6}$ |
| G | $f_{G,1}$ | $f_{G,2}$ | $f_{G,3}$ | $f_{G,4}$ | $f_{G,5}$ | $f_{G,6}$ |
| T | $f_{T,1}$ | $f_{T,2}$ | $f_{T,3}$ | $f_{T,4}$ | $f_{T,5}$ | $f_{T,6}$ |

   (b) What is the "consensus" sequence of this motif?

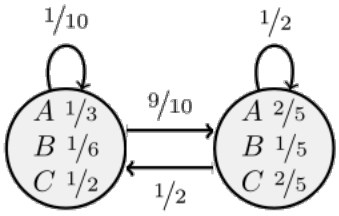
   (c) Why does the profile contain no row for the gap ("-") symbol?

   (d) Consider the following two DNA sequences of the same length:
   Sequence 1: `TGGGCA`
   Sequence 2: `CTGGCT`

      1. Which of the two sequences is more similar to the consensus sequence using the Hamming distance?
      2. Which of the two sequences fits the motif better? (Any idea of how to compute this? No need to calculate a final solution, just show how you would be able to obtain it ...)

2. **Hidden Markov Model (HMM)**

Consider the following graph representing a very simple HMM:




Let's call the two hidden states "L" (left) and "R" (right). The observable symbols which can be emitted by both states are "A", "B" and "C".

The transision probabilites $a_{LL}$, $a_{LR}$, $a_{RR}$ and $a_{RL}$, as well as the emission probabilities $e_L(A)$, $e_L(B)$, $e_L(C)$, $e_R(A)$, $e_R(B)$, and $e_R(C)$, can be obtained from the figure.

Assume the initial state probabilies to be: $P_{start}(L) = 2/10$ and $P_{start}(R) = 8/10$.

(a) Calculate the probability of having the following given hidden path $\pi$ and observed sequence $x$:

$$\pi = \texttt{R R L L R L R}$$
$$x = \texttt{B A C A C A A}$$

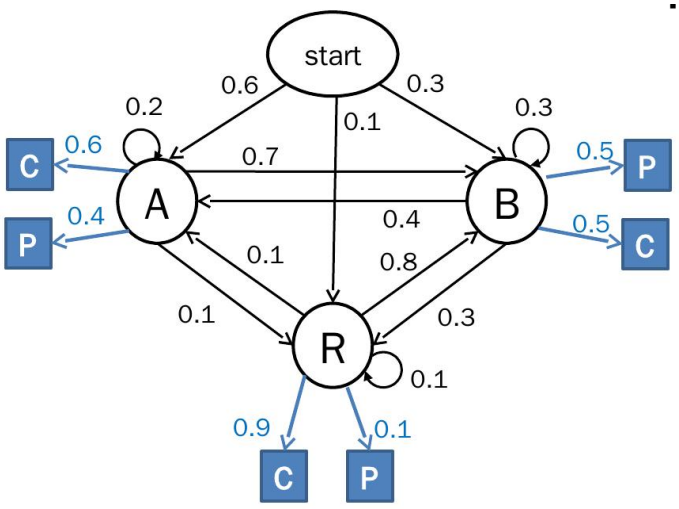
(No need to calculate the final result. Just compose the equation out of the individual probabilities to show how to compute the final probability.)

(b) How would you compute the overall probability to obtain the sequence "CAB" from this HMM? No need to calculate the final result, just show the procedure!

3. **Viterbi algorithm and most likely path**

(Note: this exercise serves to let you better understand how the Viterbi algorithm works, but the procedure is too long as an exercise for the exam; you should however know what purpose this algorithm serves and how it works!)

Consider the following HMM:




The HMM shown in the figure represents three possible hidden states ("A", "B" and "R") and two observable symbols ("C" and "P"). The **black** arrows show the initial state probabilities (from "start") and the transition probabilities between the three states. The **blue** arrows show the emission probabilities associated with the individual states.

What is the most likely hidden path to have produced the following sequence of observed symbols?

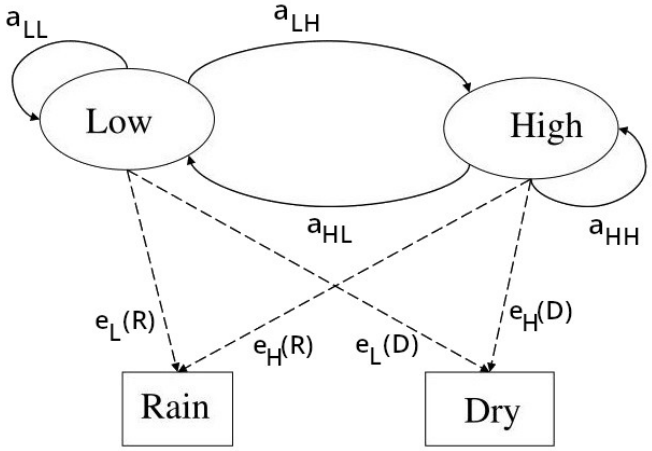
P C C P C P

Use the Viterbi algorithm to fill the following matrix and find the solution.

| Step $i$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Emission | P | C | C | P | C | P |
| A | | | | | | |
| B | | | | | | |
| R | | | | | | |

4. **Supervised learning of HMM parameters**

Consider the following HMM:




(a) Determine the maximum likelihood parameters of the HMM from the following labeled training data:

$\pi$ = L H H H L L H H H L H L H L L L H L L H
$x$ = R D D R D R R R D D R R R D D D R D D D

(b) Are there some parameters that cannot be easily estimated from this training data?